

PUNJAB RANGELAND REHABILITATION

Sector: Agriculture

Hazard addressed: Droughts

Targeted assets: Rangelands, Buffalo, Cattle, Goat, Sheep



Description: This measure involves rehabilitation of degraded rangelands through reseedling of native drought-tolerant grasses and forage species to restore vegetation cover and improve ecosystem productivity. The intervention enhances fodder availability, reduces soil erosion, improves watershed functions, and strengthens resilience of rangeland ecosystems and pastoral livelihoods against drought and climate variability.

Priority Districts: Attock, Chakwal, Jhelum, Khushab, Mianwali, Rawalpindi, and Sargodha.

TECHNICAL SPECIFICATIONS/DIMENSION



Unit of implementation: Hectare (ha) of reseeded rangeland.

Scale assumptions: Applicable in degraded rangelands vulnerable to vegetation loss, overgrazing, soil erosion, and prolonged drought conditions.

Key design parameters: Use of native drought-tolerant grass and forage species, reseedling techniques, soil and water conservation measures, grazing management systems, and community-led maintenance approaches.

Time horizon: Start implementation: 2035 onwards; ecological restoration and productivity benefits continue over long-term periods with proper maintenance and grazing control.

IMPACT



Risk reduction level: Moderate to High.

Time to effectiveness: Medium- to long-term following establishment of vegetation cover.

Expected impact: Improved vegetation cover and fodder availability, reduced land degradation and soil erosion, enhanced watershed health, and strengthened resilience against drought and climate variability.

IMPLEMENTATION CONSIDERATIONS



Enabling conditions: Community participation, grazing management plans, institutional coordination, availability of native seed species, and long-term ecosystem monitoring.

Required institutions: Forest & Rangeland Departments, Livestock Department Punjab, Environment Protection Agencies, Local Government Department, and development partners.

Barriers: Overgrazing, land tenure conflicts, limited maintenance funding, weak institutional coordination, and climate stress affecting vegetation establishment.

COST ASSUMPTIONS



Unit cost:

PKR 11,424 per hectare (~ USD 41 per hectare).

Capital Expenditure (CAPEX):

USD 2.7 million per year (2035 - 2037). Includes reseedling, site preparation, soil stabilization, fencing/protection, and restoration activities.

Operational Expenditure (OPEX):

USD 0.2 million per year (2038 - 2039). Includes maintenance, reseedling, monitoring, and grazing management activities.

CO-BENEFITS & TRADE-OFFS



Environmental: Enhanced biodiversity, carbon sequestration, improved watershed functions, and reduced desertification.

Social: Improved resilience and ecosystem services for pastoral and rural communities.

Economic: Improved grazing productivity and reduced environmental degradation costs.

Potential risks: Overgrazing pressure, poor germination/survival rates, weak maintenance systems, and delayed ecological recovery.

BENEFICIARIES



Primary beneficiaries: Pastoral communities, livestock farmers, rangeland users, environmental agencies, and rural households dependent on grazing systems.

Distributional aspects: Particularly beneficial for drought-prone and ecologically degraded rural regions vulnerable to land degradation and fodder shortages.